

Future Writing Assistants for Qualitative Research

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ABSTRACT

Qualitative analysis can be an extremely time-intensive process. Various writing assistants have been developed for qualitative analysis (QA), but they all stop short of providing a nuanced analysis of qualitative data. Future QA writing assistants could leverage large language models (LLMs) to help change this. We explore how future writing assistants using these models might benefit QA and qualitative research.

1 INTRODUCTION

Qualitative Analysis (QA) can be very time-intensive. One common approach to QA involves researchers conducting interviews to collect user perceptions of a given product. The researchers then familiarize themselves with the user responses, iteratively identifying codes or ideas within the responses, and annotating the data according to those codes. Researchers will then study the codes and responses further to generate overarching themes to then use in writing a research paper. Researchers may repeat this process multiple times before settling on a final set of themes. It has been estimated that one hour of interview data leads to approximately ten hours of analysis [6]. Researchers could easily have ten or more hours of interview data, which could mean weeks of analysis.

The corporate sphere has created a set of writing assistants, known as computer-assisted qualitative data analysis software (CAQDAS), to assist researchers with QA [3, 7, 12]. These tools primarily focus on helping researchers to annotate, organize, and visualize their qualitative data. Some, like Lexalytics, offer very basic assistance in recommending potential themes, but this functionality is based on features, like prominent noun-phrases, which may not lead to the best themes. This means that researchers are still fully responsible for coding and generating themes.

The last few years have seen drastic increases in both the abilities of large language models (LLMs) and interest in applying them to a growing array of writing tasks. Authors are using LLMs to help them brainstorm when writing fiction, summarizations, or even articles [1, 2, 10, 11]. GPT-4 was only recently released and its capabilities are still being explored, but early experimentation with it appears promising [9]. We argue that LLMs can be used to benefit researchers on several qualitative research-related writing tasks.

2 BENEFITS TO QUALITATIVE ANALYSIS

LLMs can benefit QA by cutting the time-cost for QA, quickly offering access to multiple analyses, addressing inconsistencies associated with researcher bias, and helping researchers more fully explore themes within their data.

2.1 Time Cost

First, LLMs could significantly reduce the time cost of QA. Given a set of participant responses to an interview or survey question, an LLM can be asked to automatically code the responses in a matter of seconds. Researchers rarely share their code books, which makes it difficult to assess how well model-generated codes align with human researchers, but preliminary experiments with GPT-3 and GPT-4 show that both models do show promise on this task.

Perhaps more significantly, a researcher could skip the coding process altogether. A set of participant responses to a question can be given as input to an LLM. The LLM can then be asked to generate themes and discussion based on the data. Model performance on this task appears to be improving significantly with each progressive model release. For example, GPT-4 can now simply be given a set of participant responses and then asked to generate analyses. A brief example of a GPT-4-generated thematic analysis can be found in Table 1. These examples were generated using data taken from [8]. The model was given the set of participant responses to an interview question about their experiences with physical pain as professional esports players. The model was then asked to generate a thematic analysis of the data. Table 1 is the result. We do not have room to include the original participant responses or the published discussion section, but they can be viewed at [8]. The published discussion focuses on three ideas related to physical pain, causes of pain, and prevention of pain. These ideas align closely with the themes in Table 1. The model does not include a separate theme for causes of pain, but it does discuss possible causes in the discussion portion of the themes.

We have found in our own research that GPT-3 and GPT-4 are capable of generating themes strikingly similar to those written about by human researchers in published discussion sections. The model will not always generate themes that align with those written about by humans, but in many cases the model will generate at least one theme that aligns very closely with what human researchers wrote for the exact same data. The fact that LLMs do not always align perfectly with human-generated themes also may not be an issue, given that two human researchers may choose to focus on completely different themes even for the same data. The fact that the model is *capable* of generating nearly-identical themes is encouraging.

2.2 Multiple Analyses

Table 2 contains two more examples of model-generated analyses, one using grounded theory and one using case study analysis. Again, these two model-generated analyses, though short, do follow the format we would expect to see for each of these types of analysis. They are also consistent with the data and accurately portray the ideas expressed by the participants in the study. The case study

Table 1: Example of model-generated thematic analyses for participant responses to a question about their experiences with physical pain as a result of playing esports professionally. The raw qualitative data are taken from [8].

Thematic Analysis:

1. *Back and neck pain: Several participants mentioned experiencing back and neck pain due to bad posture while playing.*
2. *Eye strain and dryness: Many participants reported eye issues such as dryness, irritation, and exposure to blue light affecting sleep.*
3. *Wrist pain and discomfort: Some participants experienced wrist pain, possibly due to excessive use of mouse or keyboard.*
4. *Fatigue and sleep disruption: A few participants mentioned gaming affecting their sleep or experiencing fatigue during the day.*
5. *Preventive measures and management: Some participants discussed taking breaks, using specific equipment, or maintaining a stretching routine to alleviate pain.*

analysis also correctly identifies each of the three participants it focuses on.

As seen in Table 2, GPT-4 can be asked to generate multiple types of analyses for the same data in a matter of seconds. It even be asked to generate a meta-synthesis of these different analyses. A researcher could use the themes from any point in this process to either compare against their own analysis or to simply take from the output and use when writing a research paper. The model can also be asked to write a basic discussion of any of these analyses, which a researcher could then edit and include in their research paper. Researchers could use this as a way to more quickly explore the themes and ideas within their data and to more quickly write research papers.

There may be frameworks and approaches for which LLMs are less-effective or for which it could be argued the use of LLMs would be inappropriate. Content analysis is one example, given that it typically involves some level of quantification and LLMs have a tendency to struggle with counting. GPT-4 is *capable* of generating something that looks like content analysis, but it would almost certainly have incorrect counts. Grounded theory may be another example. As evidenced by Table 2, GPT-4 is again *capable* of generating something consistent with the data that also appears consistent with what we might expect for a grounded theory analysis. However, one could also argue that, given the level of researcher involvement typically expected in grounded theory, that such an analysis is categorically invalid. Such a debate is beyond the scope of this three-page position paper. We simply offer that LLMs are able to generate something appearing strikingly similar to the end result we would expect for such an analysis and that such output could be beneficial to researchers wishing to quickly explore as many themes related to their data as possible.

2.3 Theme Exploration

LLMs can add perspective and potentially provide an extra check that researchers have reached saturation (the point at which no new themes can be found). Model temperature can be used to mitigate this process. We found that using a lower temperature with GPT-3 typically provides a quick overview of the main surface ideas within the data. Alternatively, using a mid-range temperature tends to generate results that are still consistent with the data, while offering a wider range of themes. Researchers could potentially use a lower temperature in the early stages of QA, to get an overview of their data and see how well their own initial conclusions match the model. A higher temperature could then be used as a way to

explore other possible themes they had not considered on their own, as a check that they have reached saturation. Multiple runs with the model at a higher temperature can yield different sets of themes. Multiple runs could be used for theme exploration. This process could help researchers identify themes they may not have identified on their own that they may then wish to include when writing a research paper for that qualitative data.

2.4 Researcher Bias

LLMs may also be able to address challenges associated with researcher bias. In QA, researchers are strongly encouraged to be aware of and transparent about their backgrounds and biases [4], however there is no way to check that they do so. While LLMs are not without their own biases [5], unlike human researchers, LLM biases can be studied and, to some extent, quantified. In some cases the model training data, i.e. background, can also be studied or adjusted.

3 BENEFITS IN OTHER ASPECTS OF QUALITATIVE RESEARCH

LLMs can be used by researchers at multiple stages of the qualitative research process to help prepare text materials for qualitative studies, particularly those involving semi-structured interviews or surveys, which require researchers to prepare a text outline of questions to be asked.

3.1 Artifact Generation

LLMs can also be used to help with artifact generation by helping researchers generate interesting questions for a given research topic. This could help researchers plan their studies or get extra perspective during the artifact design process. This could be particularly useful when starting out in a new area of research, when a researcher may not know the important questions to ask. Qualitative studies can be expensive to run. A writing assistant that generates potential questions for use on an artifact could serve as an added check that a researcher has looked at a problem from multiple angles and explored the different questions that should be asked. This could help prevent cases in which a researcher realizes later that there were better questions to ask, but they cannot afford to rerun the study.

Table 2: Examples of model-generated analyses for participant responses to a question about their experiences with physical pain as a result of playing esports professionally. The original analysis and raw qualitative data are taken from [8].

Grounded Theory Analysis:

1. *Prolonged gaming sessions: Extended hours of gaming contribute to physical discomfort and pain in various forms.*
2. *Posture and seating: Inadequate seating and poor posture contribute to back and neck pain.*
3. *Repetitive strain injuries: Continuous use of gaming equipment, such as keyboards and mice, leads to wrist pain and discomfort.*
4. *Screen exposure: Extended exposure to screens causes eye strain, dryness, and sleep issues.*
5. *Coping strategies: Participants employ various strategies, including taking breaks, using specialized equipment, and maintaining exercise routines, to mitigate pain and discomfort.*

Case Study Analysis:

1. *Participant 3's case: This participant's experience of sleep disruption, fatigue, and jitteriness highlights the potential impact of esports-related pain on daily life and mental well-being. Their reluctance to disclose the cause of their pain to a healthcare professional may indicate a stigma around gaming-related health issues.*
2. *Participant 5's case: This participant's experience of overexposure to blue light demonstrates the importance of proper time management and raising awareness about the potential consequences of excessive screen exposure.*
3. *Participant 9's case: This participant's experience of wrist pain and the need to change their gaming equipment highlights the importance of ergonomic design and the potential long-term benefits of using specialized equipment to prevent or alleviate pain.*

3.2 Synthetic Response Generation

LLMs could also help researchers by generating synthetic participant responses. LLMs could be used early in a study for a researcher to gather initial synthetic data and test that their anticipated analysis processes will work for the data they plan to collect. It could also be used to model participant responses for research that would be unsafe or unethical for human participants. Synthetic response generation could also be used to help researchers anticipate ideas participants may have, so they could include questions related to those ideas when writing their interview scripts or survey questions. This could also help avoid instances in which researchers realize after the study is over they should have asked different questions.

4 CONCLUSION

Qualitative research can involve a lot of writing, both in preparing materials for use in a study and in conducting QA while preparing a publishable research paper. Future QA writing assistants could incorporate LLMs to help researchers with these writing tasks. LLMs show promise in being able to conduct QA in alignment with human analyses in significantly less time, offering both codes and outright analyses. Because LLMs are able to do this quickly, they could be used to help researchers test multiple QA approaches for their data, potentially allowing researchers to conduct a more thorough analysis. This could lead to deeper and more thorough discussions in research papers. This could also help researchers mitigate the extent to which their own biases influence their results.

LLMs could also assist researchers in preparing materials for use in interviews and surveys. Researchers will often prepare a script or set of questions to ask in semi-structured interviews or on surveys. LLMs could help researchers explore relevant questions they may not have thought of on their own. LLMs are capable of generating synthetic participant responses to survey and interview questions. Researchers could use this as a way to explore potential ideas and responses they might expect to receive from participants in their study. This could help them prepare better questions related to those ideas.

Ultimately, an LLM could be incorporated into a writing assistant to help researchers at multiple stages of qualitative research.

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